

| ASSESSMENT 1 BRIEF            |                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Subject Code and Title</b> | NDS203 Networking and Database Systems                                                                                                                                                                                                                                                                                                                          |
| <b>Assessment</b>             | Database Project                                                                                                                                                                                                                                                                                                                                                |
| <b>Individual/Group</b>       | Individual                                                                                                                                                                                                                                                                                                                                                      |
| <b>Length</b>                 | Database creation and manipulation scripts (small dataset)                                                                                                                                                                                                                                                                                                      |
| <b>Learning Outcomes</b>      | <p>The Subject Learning Outcomes demonstrated by successful completion of the task below include:</p> <p>c) Implement database design and modelling concepts focusing on system efficiency, normalization, and data integrity.</p> <p>d) Apply database querying and manipulation techniques prioritising querying strategies and performance optimisation.</p> |
| <b>Submission</b>             | Due by 11:55pm AEST/AEDT Sunday end of Module 4.                                                                                                                                                                                                                                                                                                                |
| <b>Weighting</b>              | 30%                                                                                                                                                                                                                                                                                                                                                             |
| <b>Total Marks</b>            | 100 marks                                                                                                                                                                                                                                                                                                                                                       |

### Assessment Task

Demonstrate your understanding of database creation and maintenance by creating tables and running queries using the supplied dataset.

Please refer to the Task Instructions for details on how to complete this task.

### Context

In this assessment, you will be provided with a large dataset. You are required to design a database in MySQL Workbench that conforms to the design concepts taught during the lectures, input the data and print out multiple outputs using queries

### Instructions

An excel spreadsheet named 'Game World Data.xlsx' is provided with this brief which contains the dataset needed to populate your database.

The dataset defines a small game world of NPCs. These NPCs are grouped together by different factions, can give out quests and train others in certain skills. The database also shows the location and region the NPC can be found in this world.



### **Task 1: Design the database**

The first task is to design the database in MySQL Workbench by creating tables to store this data. You need to ensure that:

- Syntax for creating tables is correct.
- Multiple tables are created to show each relationship in the database.
- Correct relationships are identified.
- There is proper use of primary keys.
- There is proper use of foreign keys.

You should use a single SQL script to create the database that can be run once in its entirety without error.

### **Task 2: Input all the data**

Once the tables are created, you should write another SQL script to populate your database by extracting the data from the provided excel spreadsheet (not using an excel reader).

You should ensure that the whole SQL script need only be run once, after the script from Task 1 has been executed, in its entirety without error.

### **Task 3: Queries and Mutations**

After the database is fully populated, you should create a new SQL script **for each** of the following queries to generate the output defined as display columns.

- **Query 1:** Print all skills that can be learned in this database.
  - Display Columns: Name and Type.
- **Query 2:** Print all Female NPCs.
  - Display Columns: First Name and Last Name.
- **Query 3:** Print all members of the 'Rogues Guild' in ascending order.
  - Display Columns: First Name, Last Name, Gender and Faction Code.
- **Query 4:** Print all Quests given from NPCs in the location 'Arcane Capital'.
  - Display Columns: Quest Name, NPC First Name.
- **Query 5:** Print all Quests that are given out by 'Warriors United' in a 'Large Scale City'.
  - Display Columns: Quest Name, NPC First Name.
- **Query 6:** Print the number of quests given in each Location.
  - Display Columns: Location Name, Number of quests.
- **Query 7:** Add a one-to-many relationship between the 'Factions' and 'Skills' to the already existing database. Call this new relationship 'Favourite Skill' and update each faction with the following skills;
  - School of Magicians: Fireball.
  - Warriors United: Lunging Strike.
  - Rogues Guild: Steal.



- After adding the above requirements for Query 7, print each Faction with its Favourite Skill.
  - Display Columns: Faction Name, Favourite Skill Name.
- **Query 8:** Change the data type of the field 'Gender' to CHAR() and update every NPC to hold the values 'Male' and 'Female' instead of 'M' and 'F' respectively using a single update call. Then show all NPCs.
  - Display Columns: First Name, Last Name and Gender

You should ensure each SQL script, when run in sequence, executes in its entirety with no errors.

### Referencing

It is essential that you use appropriate APA style for citing and referencing research. Please see more information on referencing here <https://library.torrens.edu.au/academicskills/apa/tool>

### Submission Instructions

Submit a Zip file via the Assessment link in the main navigation menu in NDS203 Networking and Database. The Learning Facilitator will provide feedback via the Grade Centre. Feedback can be viewed in My Grades.

The Zip file should contain the following SQL (.sql) files:

- Task 1 – Database creation.
- Task 2 – Database population.
- Task 3 – All SQL files labelled as Query1, Query2 etc.

Name the zip file in the following format NDS203\_Assessment3\_LastName\_FirstName.zip

Before you submit your assessment, please ensure you have read and understand the conditions outlined in the Academic Integrity Code Handbook. If you are unsure about anything in the Handbook, please reach out to your Learning Facilitator.

### Academic Integrity Declaration

I declare that except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at <http://www.torrens.edu.au/policies-and-forms>

I am aware that I need to keep a copy of all submitted material and their drafts, and I will do so accordingly.

### Assessment Rubric

| Assessment Attributes                                              | Fail<br>(Yet to achieve minimum standard)<br>0-49%                                           | Pass<br>(Functional)<br>50-64%                                                                                         | Credit<br>(Proficient)<br>65-74%                                                                                                                                                                              | Distinction<br>(Advanced)<br>75-84%                                                                                                                                                                                | High Distinction<br>(Exceptional)<br>85-100%                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task 1: Design Database<br><br>Percentage for this criterion = 50% | 25%                                                                                          |                                                                                                                        |                                                                                                                                                                                                               |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                        |
|                                                                    | Unable to see evidence of clear database design of tables with appropriate names and fields. | Demonstrate a satisfactory database design in single SQL file with respect to; design for the tables including naming. | Demonstrate a functional database design in single SQL file with respect to; design for the tables including naming. Further demonstrate a satisfactory database design with respect to 3rd Normal Form (3NF) | Demonstrate a proficient database design in single SQL file with respect to; Appropriate design for the tables including naming. Further demonstrates a proficient database design by using appropriate datatypes. | Demonstrate a outstanding database design in single SQL file with respect to; design for the tables including naming. Data provided separated into smaller files and SQL script executes in its entirety without any errors or warnings, using appropriate datatypes, including attribute constraints such as NOT NULL |

|                                                                      | 25%                                                                                                                                                           |                                                                                                                                                           |                                                                                                                                                                                                                                                  |                                                                                                                                                                                              |                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      | The SQL file fails to generate a database by executing the file once in its entirety.                                                                         | Appropriate field names. Executes smoothly with a single file run once in its entirety.                                                                   | Data is not repeated where possible, foreign keys and composite entities used to mediate complexity                                                                                                                                              | Appropriate use of defining fields with NOT NULL                                                                                                                                             | Proficient use of primary and foreign keys to define relationships between tables                                                                                                                                                             |
| Task 2: Input Data<br><br><b>Percentage for this criterion = 30%</b> | Unable to see evidence of data being input into the database<br>OR<br>The SQL file fails to populate the database by executing the file once in its entirety. | Demonstrates satisfactory implementation to populate the database using majority of the data provided.                                                    | Demonstrates functional implementation by populating the database using the data provided.<br>Further demonstrates satisfactory implementation by using appropriate identifiers as primary keys.<br>Uses appropriate identifiers as primary keys | Demonstrates proficient implementation by populating the database using the data provided.<br>Further demonstrates proficient implementation by creating input for ALL of the data provided. | Demonstrates outstanding implementation by populating the database using the data provided.<br>Further uses appropriate identifiers as primary keys<br>All of the input data is correctly entered without any spelling or grammatical errors. |
| Task 3: Queries<br><br><b>Percentage for this criterion = 20%</b>    | Unable to see at least FOUR queries in separate SQL files that output the correct result when each SQL file is run once in its entirety.                      | Satisfactory implementation of at least FOUR queries in separate SQL files that output the correct result when each SQL file is run once in its entirety. | Satisfactory implementation of at least SIX queries in separate SQL files that output the correct result when each SQL file is run once in its entirety.                                                                                         | Satisfactory implementation of ALL queries in separate SQL files that output the correct result when each SQL file is run once in its entirety.                                              | In addition to the requirements of distinction, the queries are written with a concise and optimal implementation.                                                                                                                            |